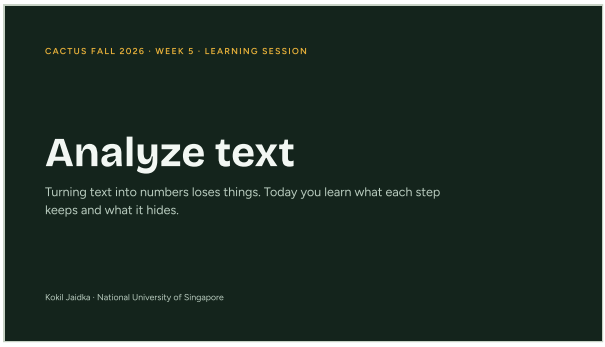
CACTUS Week 5: Analyze Text



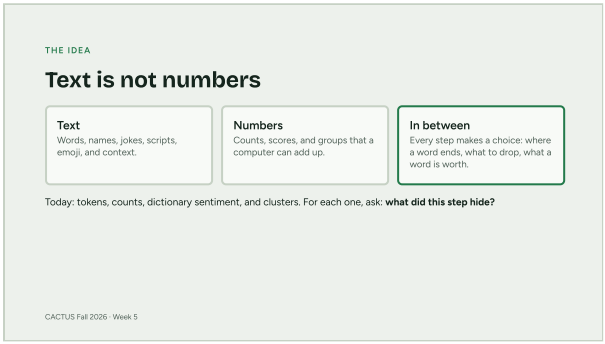
Slide 1

TIME 1 min (0:00)

SAY Welcome back. Last week you collected data. Today we turn text into numbers, and we keep asking one question: what did this step hide?

DO Welcome people. Post the course site link. Ask who has already tried a text analysis tool before (hands or chat).

NOTE This is a 60-minute session. The coding happens in the coding session.



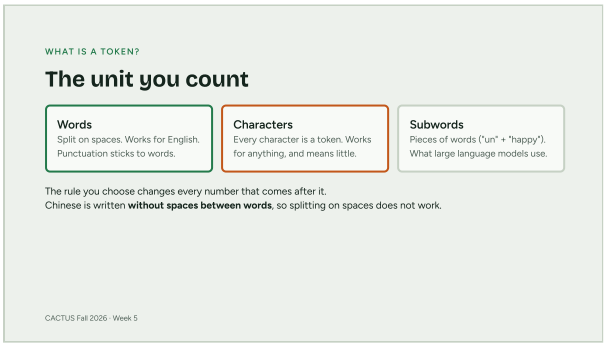
Slide 2

TIME 4 min (0:01 to 0:05)

SAY Text is words, names, jokes, scripts and context. A computer wants numbers. In between there are choices: where a word ends, what to drop, what a word is worth. Each choice changes the result.

DO Read the three cards. Say that today has four steps: tokens, counts, dictionary sentiment, clusters. For each, we ask what it hid.

IF ASKED about LLMs say Week 6 uses an AI to label texts and measures how well it agrees with humans. Today is the older, simpler methods, which you must understand to judge anything newer.



Slide 3

TIME 6 min (0:05 to 0:11)

SAY A token is the unit you count. Words, characters, or pieces of words. The rule you choose changes every number after it. English has spaces. Chinese does not put spaces between words.

DO Ask the room: how many words are in the Chinese sentence on the next slide? Let two people answer before you show it.

IF ASKED about subwords large language models cut text into pieces, so what the model sees is not whole words. You do not need more than that today.

WATCH FOR Do not make this about Chinese being hard. Say that every language has cases where a tokenizer guesses.



Slide 4

TIME 6 min (0:11 to 0:17)

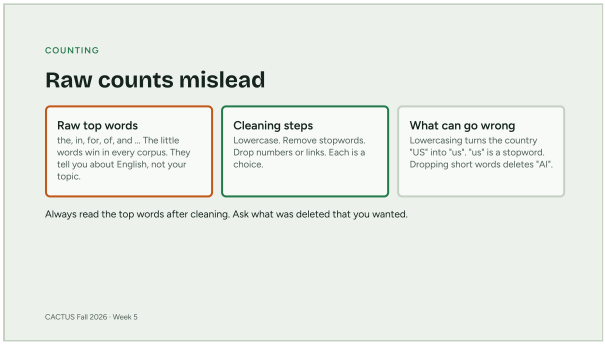
DO (live demo) Run the five practice-page texts through splitting on spaces and through jieba in a notebook cell, or read the table on the slide. Point at the English row, the Chinese row, and the mixed row.

SAY Splitting Chinese on spaces gives one token. jieba finds words. But look at 寫功課: jieba cut it as 寫功 and 課, which is wrong. It also splits English punctuation. No tokenizer is right everywhere.

DO Ask: which tokenizer would you trust on your own text, and how would you check? (Read a sample of the tokens.)

WATCH FOR People who leave thinking jieba is correct. Say it is a good start, and you read its output.

NOTE jieba must be installed first (it takes under a minute). Do this before the session.



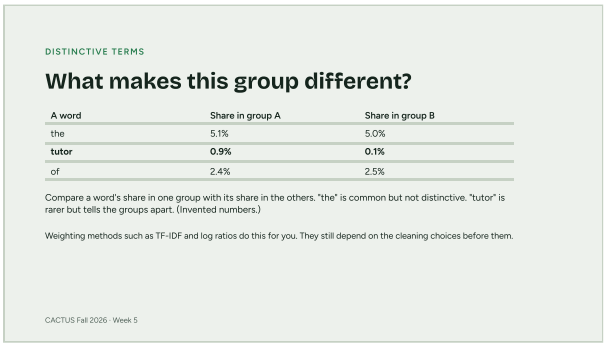
Slide 5

TIME 6 min (0:17 to 0:23)

SAY Count the words in any corpus and the little words win: the, in, for, of. They tell you about the language, not your topic. So we clean: lowercase, remove stopwords. Each step is a choice, and each can delete something you wanted.

DO Say two examples. Lowercasing turns the country US into us, and us is on most stopword lists, so the country disappears. Dropping words under three letters deletes AI. Say the practice page lets them try both.

WATCH FOR People who treat cleaning as automatic. Say: after cleaning, always read the top 20 and ask what is missing.



Slide 6

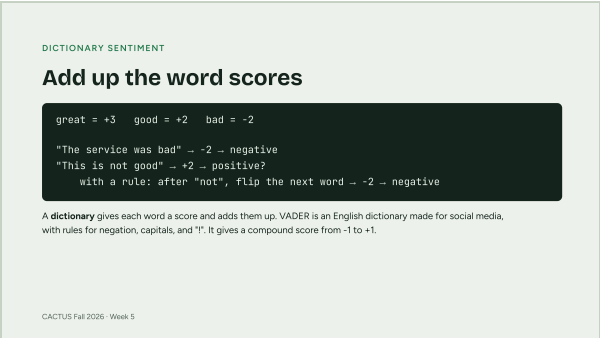
TIME 7 min (0:23 to 0:30)

SAY Raw counts show what is common everywhere. To see what makes a group different, compare a word's share in this group with its share in the others. The is common in both, so it is not distinctive. A rarer word used much more in one group is.

DO Read the table. These numbers are invented. Say that TF-IDF and log ratios are ways of doing this, and the notebook has a simple one.

IF ASKED which to use say it depends on the question, and that all of them still rest on the cleaning choices before them.

WATCH FOR Treating the distinctive list as the truth. It is a list of candidates to read in context.



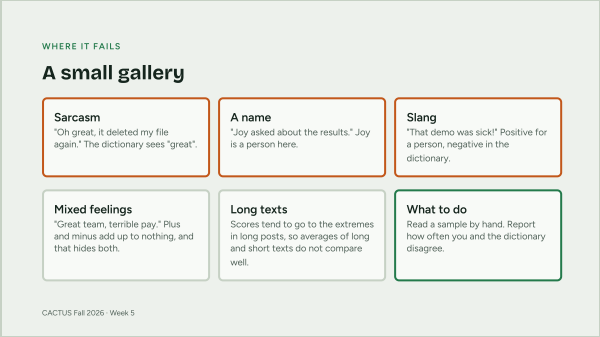
Slide 7

TIME 5 min (0:30 to 0:35)

SAY A sentiment dictionary gives each word a score and adds them up. This is a toy dictionary. Not is the classic problem: this is not good adds up to positive, unless the dictionary has a rule that flips the next word. VADER is a real English dictionary made for social media, with a few such rules. It gives a compound score between minus one and plus one.

DO Read the two examples. Say the practice page lets them edit a toy dictionary and find the texts no edit can fix.

IF ASKED which cut-offs say plus or minus 0.05 is a common convention, and that it is a choice.



Slide 8

TIME 7 min (0:35 to 0:42)

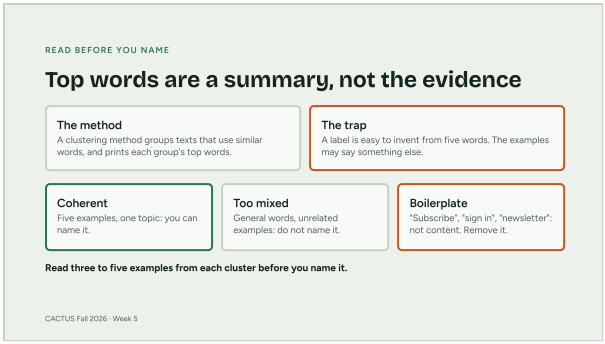
SAY Here is where a dictionary fails. Sarcasm: oh great, it deleted my file. A name: Joy. Slang: sick. Mixed feelings: great team, terrible pay adds up to near zero, which hides both. And in long texts the scores tend to go to the extremes, so averages across long and short texts do not compare well.

DO Ask the room for one more example from their own language or field. Take two.

SAY (important) the way to handle this is not to throw it away. Read a sample by hand and report how often you and the dictionary disagree.

WATCH FOR Do not say what an English dictionary does to Chinese text. Students find that in Exercise 2d, and it is one of the three planted mistakes in Exercise 3.

WATCH FOR People who conclude that sentiment tools are useless. Say they are useful when you know their limits and report them.



Slide 9

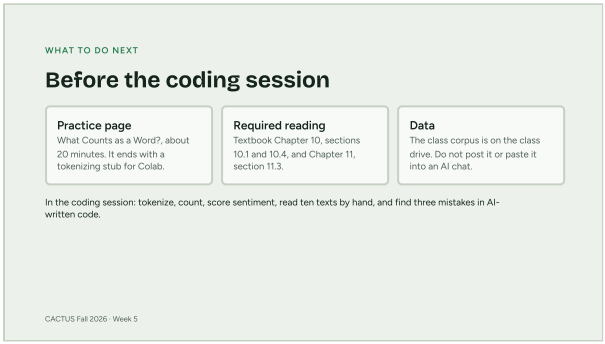
TIME 10 min (0:42 to 0:52)

SAY A clustering method groups texts that use similar words and prints each group's top words. It is easy to invent a name from five words. The examples may say something else.

DO Go through the three cards: one coherent cluster, one too mixed, one that is boilerplate. Then ask the room to say what they would do with each. Do not rush; this is the main point of the session.

SAY Read three to five examples from each cluster before you name it. If they do not fit one label, say so. If the cluster is page boilerplate, remove it and say you did.

WATCH FOR Students who want to skip reading because there are many clusters. Say reading ten examples takes five minutes.



Slide 10

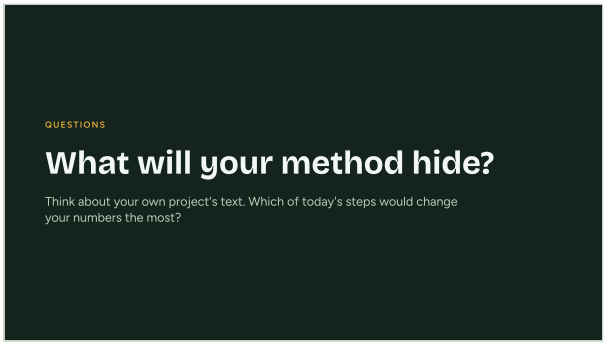
TIME 4 min (0:52 to 0:56)

SAY Before the coding session: the practice page (about 20 minutes), and the required reading (textbook Chapter 10, sections 10.1 and 10.4, and Chapter 11, section 11.3).

DO Post the practice page link and the course site link. Tell people where to get the class data (the class drive) and how to handle it: course use only, do not post it, do not paste it into an AI chat, do not try to identify anyone.

SAY In the coding session there is no lecture. Tokenize, count, score sentiment, read ten texts by hand, and find three mistakes in AI-written code.

SAY (important) the Reddit posts are real people's words. Read them as data and with care.



Slide 11

TIME 4 min (0:56 to 1:00)

SAY Think about your own project's text. Which of today's steps would change your numbers the most? Write one sentence in your project card.

DO Take questions. If you do not know the answer, say so and write it down.

CLOSE Thank everyone. Say the next session is the coding session.
